The empirical column recurrences for the table OEIS A264128

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Abstract

OEIS A264128 is a two-dimensional table $T(n,k)$, and it carries a block of empirical recurrences contributed by R. H. Hardin — one for each of its first few columns. They are true. Column k of the table is an ordinary fixed-width array count, so its rows are the vertices of a finite digraph and $T(n,k)$ is a number of walks in it; being a walk count the column is C-finite, and whether a given recurrence annihilates it is decided by a finite exact computation, with the Cayley–Hamilton theorem bounding the work. This note settles the columns $k = 1$.

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1 The table and the conjectures

OEIS A264128 is "T(n,k)=Number of (n+1)X(k+1) arrays of permutations of 0..(n+1)*(k+1)-1 with each element having index change +-(.,.) 0,0 1,-2 or 2,-1.". It has offset 1, is read by antidiagonals, and begins

$$1, 2, 2, 4, 9, 4, 8, 45, \ \ldots$$

The entry carries the following block:

Empirical for column k:
k=1: a(n) = 2*a(n-1)
k=2: a(n) = 4*a(n-1) +a(n-2) -4*a(n-3) +16*a(n-4) +4*a(n-5) +a(n-6) -4*a(n-7) -a(n-8)
k=3: a(n) = 10*a(n-1) for n>5
k=4: [order 8] for n>11
k=5: [order 32] for n>34
k=6: [order 95] for n>99

As of the "Last modified" line on the live entry (Nov 10 2015 11:03:29, revision 6) these are still recorded as empirical, and nothing on the entry records any of them as settled.

2 A column of the table is a walk count

Fix k. The entry defines $T(n,k)$ as the number of arrays of a shape in which one dimension is n and the other is determined by k, subject to a condition imposed on every 2×2 subblock — a condition local to two consecutive rows and two consecutive columns. Substituting the fixed value of k into the entry's own wording turns the definition into an ordinary one-parameter array count, and for such a count the rows are the vertices of a finite digraph G_k: $r \to s$ is an edge exactly when placing s under r satisfies the condition in every column pair. An admissible array is then a walk, so with M_k the adjacency matrix of G_k,

$$T(n,k) \ = \ c_k \, \mathbf{1}^{\top} M_k^n \mathbf{1}$$

for the constant c_k that the entry's name supplies (a factor such as $\frac{1}{2}$ or $\frac{1}{4}$ where the name says so, and 1 otherwise).

Lemma 1. *Let $q(t) = t^r - \sum_i c_i t^{r-i}$ be the characteristic polynomial of a conjectured recurrence of order r for column k. Then for every $n \geq r$,*

$$T(n,k) - \sum_i c_i \, T(n-i,k) \;=\; c_k \, \mathbf{1}^\top M_k^{n-r} q(M_k)\mathbf{1},$$

so the recurrence holds from some point on if and only if $\mathbf{1}^\top M_k^j q(M_k)\mathbf{1} = 0$ for every $j \geq 0$; and by the Cayley–Hamilton theorem it is enough to check $j < S_k$, where S_k is the number of vertices of G_k.

Proof. Substitute the walk expression and factor M_k^{n-r} out of $M_k^n - \sum_i c_i M_k^{n-i}$; the scalar c_k is nonzero. For the bound, $M_k^{S_k}$ is an integer combination of $I, M_k, \ldots, M_k^{S_k-1}$, so each $\mathbf{1}^\top M_k^j q(M_k)\mathbf{1}$ with $j \geq S_k$ repeats that combination of earlier values. □

3 The computation, column by column

line	order	vertices S	proved for	terms checked
column k= 1	1	256	> 1	9

In each case the vector $w = q(M_k)\mathbf{1}$ was formed by r matrix–vector products in exact integer arithmetic and $\mathbf{1}^\top M_k^j w$ evaluated for $j = 0, 1, \ldots,$ again exactly; every one of those integers vanishes from the stated point onwards and the run continued until the working vector was identically zero, which settles all larger j at once.

Column k= 1

The sequence is "Number of (n+1)X(2) arrays of permutations of 0..(n+1)*(2)-1 with each element having index change +-(.,.) 0,0 1,-2 or 2,-1.". Its rows are the vertices of a digraph on $S = 256$ vertices, edges being the admissible consecutive pairs, and the count is $\mathbf{1}^\top M^n \mathbf{1}$ up to the constant factor in the entry's name. The conjectured recurrence

$$a(n) \;=\; 2\,a(n-1)$$

holds from index $1 + 1$ on.

4 Verification

Three checks, each independent of the algebra above.

First, each model was compared against the entry's own published values for its line, taken both from the antidiagonal DATA read in either orientation and from the "Table starts" block, and agrees at every available term. This is what ties a model to the table: substituting the fixed index into the entry's wording is a reading of English, and a wrong reading gives a different digraph and different counts. Across the lines settled here the model reproduces 9 published values, 16 decimal digits in all, every one of them exactly; a misreading of the entry would have to reproduce all of them by accident.

Second, each conjectured recurrence was evaluated on those published values in exact integer arithmetic, with no matrices involved, and holds wherever the proved range and the data overlap.

Third, the annihilation test of Lemma 1 was carried out in exact integer arithmetic rather than on a sampled prefix.

Remark 2. A row of the table is a column of the transposed table: fixing n and letting k vary gives a count of arrays of fixed height, which is the same kind of object with the two dimensions exchanged. Both are settled here by the same argument. Only the lines the entry states explicitly, and for which enough published values exist to run the first check above, are claimed.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A264128.

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[4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.